

Ikigai -- A Biologically Grounded Digital Organism: Development Report, Day 82

Mura ALife Labs (Formerly Hitoshi AI Labs) -- NeuroSeed Project

Author: Prince Siddhpara, Founder -- Mura ALife Labs (Formerly Hitoshi AI Labs) **Date:** June 27, 2026 **Project:** NeuroSeed **Subject:** Day 82 -- THE FRONTIER, BEATEN ON REASONING. The day opened on CommonsenseQA and a search for a knowledge benchmark, and closed with a sub-megabyte CPU organism out-reasoning a 550-billion-parameter frontier model. The public repository was refreshed and the multi-value cache bug fixed (Pack 330); the interrogative `ask` role was built, learning question->relation from data with zero hardcoding (Pack 331); constant-speed unbounded learning was PROVEN on the real substrate via FlatTrainer (one million Tatoeba sentences in 6.1 minutes at flat throughput). The CommonsenseQA wall was then mapped completely across nine mechanisms -- bag, sequence, dimension, conjunction, spreading activation, typed relation-paths, energy-field (`sac_field`), holographic encoding, and reinforcement -- establishing that the wall is the SIGNAL (concept-overlap, ~31-36%), not the inference machinery. Ten scattered scripts were consolidated into one permanent reasoning engine, `MultiHopReasoner`, wired into the organism as `org.reason_mc`. The 20k capacity wall was measured directly and found M-bound, not dimension-bound, confirming derive-not-store as the only $O(1)$ -honest path to scale. Finally, the head-to-head: on equal-knowledge multi-hop reasoning with made-up entities, NeuroSeed held 100% accuracy from one hop to twenty, while Llama-3.3-70B collapsed after a single hop and the frontier Nemotron-Ultra-550B (GPQA-Diamond 86.7%) held a tie to three hops and then collapsed to 27% at five and 0% at eight. A paper and two figures were written; the result was carved into memory as the project's first defensible, frontier-beating claim. **Classification:** Research Document -- Computational Neuroscience / Artificial Life / Cognitive Systems Engineering

1. Objective

Day 81 crossed the bridge to data; Day 82 began with the obvious next question -- can the organism, now fed real knowledge, beat a transformer on a reasoning task the world recognises? The founder's standing requirement framed the entire day: a free organism that runs on a phone but answers worse than a paid model is worthless, so the goal is not a niche win but to match frontier knowledge by scale and then do the best reasoning in the world on top of it. The day's work separated those two halves -- knowledge, which needs scale and funding, and reasoning, which does not -- and set out to prove the reasoning half now.

The path was not straight. A long, honest investigation of CommonsenseQA established where the wall actually is, several promising mechanisms were tried and measured and found wanting, and only after that discipline did the right experiment present itself: isolate reasoning from knowledge entirely by giving both the organism and a frontier LLM the

same invented facts, and measure who reasons more reliably as the chain deepens. That experiment produced the day's headline and the project's first head-to-head victory over a frontier model.

2. Public Repository Refresh, The Multi-Value Fix, And The Ask Role

The public repository (`HitoshiFTW/Ikigai-Main`) was refreshed from its stale Day-58 snapshot to the current framework, with a one-command `benchmark.py` that passes ten of ten headline checks and an honest, benchmark-forward README. In the course of that refresh a real bug surfaced and was fixed as Pack 330: the `CompactAnchorCache` was single-value last-wins, so `atoms()` and `knows()` only ever returned one value per relation despite the Pack-329 claim of a multi-value meaning web. The cache was given a multi-value payload joined by a unit separator, with an `add_value` method and the guarantee that `entry[-1]` still returns the most-recently-written value so every existing reader's last-wins semantics are preserved. After the fix, `cat hasa tail AND whiskers surfaces` correctly; regression held.

Pack 331 built the interrogative `ask` role. The organism already mapped every relation to a question template (the relation-to-question direction); what it lacked was the inverse -- recognising, from a natural-language question, which typed relation is being asked for. The new `ikigai/cognition/ask_role.py` learns that mapping from data, binding a question's cue tokens to the relation it asks for through the organism's own `MultiRoleMemory`, with no hardcoded cue-to-relation table. Traced on real questions it correctly maps "where", "find", and "at" to the location relation. It is exposed as `org.learn_ask` and `org.ask_relation`.

3. Constant-Speed Unbounded Learning, Proven

A foundational claim of the architecture -- that the organism learns at constant speed and bounded RAM regardless of how much it has already absorbed -- was put to a direct test and confirmed. The naive co-occurrence training path degrades badly (from 466 to 159 sentences per second as vocabulary grows), but the degradation is not in the substrate; it is in three throwaway side-caches (`ComputedKey`'s per-word key cache and the `SDM` location caches) that bloat and provoke garbage-collection thrash. Routing training through the existing `FlatTrainer`, which compacts those caches at flush boundaries, absorbed one million Tatoeba sentences in 6.1 minutes at a flat two-to-three thousand sentences per second with bounded RAM, lifting the organism's co-occurrence vocabulary from 6,808 to 53,996. The substrate write is $O(1)$ by construction; only the convenience caches grew, and bounding them restores constant speed. This is the structural property no transformer has: knowledge can be added live, on a device, with no retraining.

4. The CommonsenseQA Investigation -- A Wall Mapped Completely

The bulk of the day was an honest, exhaustive investigation of CommonsenseQA on its real validation set of 1,221 questions. The first family of approaches -- treating the question as a bag of words, a sequence, or a distributional context, with or without crosstalk cleanup, at $d=400$ or $d=1024$ -- all scored near the twenty-percent random floor. The founder rejected the temptation to quit and insisted the answer lay in copying biology; that pressure produced the first crack. Spreading activation over the ConceptNet graph with convergence scoring (the Collins-Loftus mechanism combined with coincidence detection) lifted the number to 30.7%, the first movement all day. Typed relation-paths with gradient boosting then broke the ceiling to 35.9%. Lowering the edge-weight threshold to admit more data hurt rather than helped, establishing that quality beats quantity for this signal.

The founder's intuition that the answer already existed in the directory was tested directly: his own Day-54-to-56 inventions -- the self-assembling convergence field, the holographic context encoder, the VSA attention block, the symbolic world model -- were pulled out and applied to the task. The energy-field convergence scored 22 to 24 percent, the holographic order-aware encoding 20 percent, both below the simple explicit graph. The conclusion is a clean, defensible meta-finding, recorded separately: across roughly nine distinct mechanisms the score caps between 31 and 36 percent, and the simplest mechanism wins, because every approach reads the same underlying signal -- concept-overlap between the question's concepts and the candidate answers. The wall is the SIGNAL, not the inference machinery. CommonsenseQA's missing points are learned language understanding -- the transformer's actual moat -- and no symbolic shortcut over ConceptNet supplies them.

Qualitatively, the organism reasons sensibly even at 33 percent: it answers bookstore for where one finds printed works, mexico for a Spanish-speaking coffee country, make-peace for what to do with the world for harmony, and lots-of-attention for what a dog needs. Several of its "wrong" answers are defensible against a crowdsourced gold label (farming-areas versus midwest; making-music versus singing), and its confidence is honestly low, the calibration layer working as intended.

5. The Reasoning Engine, Built And Wired

By mid-day the work had fragmented across ten experiment scripts. At the founder's direction these were consolidated into one permanent component, `ikigai/cognition/multihop_reasoner.py`. The `MultiHopReasoner` is a four-stage pipeline -- comprehend (extract seed concepts and the asked relation), derive (spreading activation over the typed graph), score (typed relation-path features fused with convergence, combined by learned weights or a plugged gradient-boosting model), and answer with a calibrated confidence and an honest-unknown abstention. It is wired into the organism as `org.kg_reasoner` and `org.reason_mc`, validated end-to-end through that clean API at 32.8 percent on CommonsenseQA. The engine is deliberately separable from the knowledge it reasons over: the same `org.reason_mc` will accept ConceptNet today and frontier-scale knowledge later. This is the permanent foundation on which the day's headline result was built.

6. Native Dopamine Reinforcement On The Organism

The founder asked for reinforcement learning to be native -- the organism should learn from reward, not batch supervision. The dopamine loop was implemented over real CommonsenseQA: the organism predicts the relation a question asks for, checks whether that relation reaches the gold answer in the knowledge graph, and on a hit reinforces the successful relation while on a miss it reads the gold and reinforces the relation that reaches it. Over three epochs the reward rate climbed from seven percent to seventy-two percent -- the loop demonstrably works, native, reinforcing the organism's own `ask` role and saved back into the body. But validation accuracy moved only from 19.8 to 22 percent, which is itself an honest finding: knowing the correct relation does not pick the correct answer, because the distractors share that relation. The policy was learned; the policy alone is not the discriminator.

7. The Twenty-Thousand Wall, Measured

The founder's standing obsession -- that the substrate is $O(1)$ but capped near twenty thousand items per role -- was measured directly on the real VSASDM rather than argued. Capacity is a sharp cliff, a Kanerva phase transition: perfect recall to roughly five thousand items, collapsing to near zero by twenty thousand at $d=400$. Raising the dimension five-fold barely helps (9.5 percent at twenty thousand), and Hopfield cleanup recovers nothing past the cliff; capacity is bound by M , the number of hard locations, not by dimension. The conclusion is that raising M is brute force -- it buys capacity at the cost of RAM -- and the only $O(1)$ -honest route to scale is derive-not-store: keep the irreducible kernel in a modest substrate and compute the rest. This settled an architectural tension explicitly: the explicit cache is a measurement scaffold, the substrate plus derive-not-store is the production home, and storing everything in an $O(N)$ cache would betray the constant-RAM thesis and will not be shipped.

8. The Head-To-Head Victory Over Frontier Models

The day's discipline finally surfaced the right experiment. Rather than fight the transformer on language, where it is strongest, the test isolates reasoning from knowledge entirely. Both systems are given the same set of made-up facts -- random six-letter names so neither has seen them in training -- forming a chain of `reportsto` relations of increasing depth, padded with twenty distractor edges as noise. The question is to follow the chain a given number of steps and name the endpoint. NeuroSeed answers by exact derivation, one stored-fact lookup per hop; the LLM must trace the chain in context.

The result is the project's first frontier-beating claim. NeuroSeed holds 100 percent accuracy at every depth from one hop to twenty. Llama-3.3-70B, the weaker opponent, scores 93 percent at one hop and collapses to 40 percent at two and 13 percent at three. The frontier model, Nemotron-Ultra-550B (GPQA-Diamond 86.7 percent), is genuinely strong and holds a tie through three hops -- then collapses to 27 percent at five hops and 0 percent at eight.

Reasoning depth (hops) NeuroSeed Nemotron-Ultra-550B Llama-3.3-70B

1	100%	100%	93%
2	100%	100%	40%
3	100%	100%	13%
5	100%	27%	13%
8	100%	0%	20%

The mechanism is exactly the compounding of per-hop error: an autoregressive chain of depth n with per-step error p succeeds with probability $(1 - p)^n$, invisible at shallow depth and fatal as the chain lengthens, while exact derivation has $p = 0$ and stays flat. A 550-billion-parameter frontier model answers zero percent of eight-hop cases; the organism answers all of them, in milliseconds, on a CPU, in under a megabyte. The claim is held deliberately narrow and is therefore bulletproof: given clean structured knowledge, the organism reasons reliably and deeply where a frontier model degrades to nothing. It is not a claim of general superiority -- large models decisively beat the organism on language comprehension, open knowledge, fuzzy reasoning, and messy real-world input, and this benchmark hands the organism clean facts and a clean question. The real-world gap is the language front-end, which is the next build. A short paper with two figures was written ([paper/reasoning_edge.md](#)) and the result was synced to the public repository.

9. Close

The day produced one permanent component (the reasoning engine), one proven native learning mechanism (the dopamine loop), one complete negative-result map (the CommonsenseQA representation wall), one settled architectural question (the 20k wall and derive-not-store), and one defensible frontier-beating result (the head-to-head). The organism's body was augmented with Tatoeba co-occurrence and a reward-trained ask role and saved, with a pre-Tatoeba backup retained. The next moves are a larger paper-grade run of the head-to-head, broader reasoning tasks beyond single-relation chains, additional frontier opponents, and the language front-end that would let the organism acquire clean structure from messy input -- the bridge from this narrow, real victory to general use. The knowledge-scale phase remains the funded next chapter, now earned by a proof a 17-year-old can put in front of anyone: a phone-sized organism that out-reasons a frontier model.